

# Response letter

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## Outline

In their letter, Bogdziewicz et al. say that the reproduction synchrony among individuals—the masting—has to be controlled by the solstice. It is explicitly one of the hypotheses our paper considers, and we do not believe our results invalidate it. We’re sorry if something in our paper may have led them to believe the opposite. Similarly, we would like to clarify that we did not address the risk of “*late frost damage*”, as stated by Bogdziewicz et al. Yet, we trully appreciate their interest in our work, and we are glad that it may stimulate fruitfull scientific discussion. Our results do show solstice is optimal, but also highlight some amount of variation across space, and time.

These variations raises important questions. The notion of “*temperature sensing window*”—in fact, some amount on correlation between an annual metric of seed production and daily temperature, computed over a certain moving-window—appears higher around the solstice (and may even start earlier, as shown in Fig1 in Bogdziewicz). Interestingly, the timing of this correlation varies across space (Journe et al., Fig. 1 in Bogdziewicz). However, the lack of quantification of this spatial variation—alongside an omission of possible temporal variation—make it difficult to compare their findings across latitude and with our results. Clearly, variation exists and will happen, but the extent to this variation could possibly invalidate the “*celestial starting gun*” [sic] hypothesis remains unclear. Similarly, we believe that the lack of a true quantification of the scale under which the “*remarkable spatial synchrony*” is happening is an important missing piece of this hypothesis.

Research on masting has gained a lot of interest in the last decade [cite]. Many hypotheses coexist, and may be supported by different underlying mechanisms. Maximizing synchrony is critical for maximum fitness under some economy-of-scale hypothesis [cite]. In the case where this hypothesis holds, we believe the synchrony could still results from alternative cues, not necessarily the solstice. While we agree on the distinction between reproduction and growth, the interest of our work is to consider the link between the solstice and these processes through a broader lens—both from a climatological and an eco-evolutionary perspectives. Our work did not aim at refuting any specific hypothesis. Instead, we believe it emphasizes that robust science depends on considering multiple mechanisms and alternative hypotheses—always with a particular caution about any causal links we believe we are testing.